McHenry County

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1 Getting Started — Connecting to FieldComms

http://

FieldComms is a self-contained emergency communications server running on a Raspberry Pi. The ASUS RT-BE58 Go travel router broadcasts a dedicated Wi-Fi network called **EMCOMM-NET** — the Pi itself does not broadcast Wi-Fi. Any smartphone, tablet, or laptop in range connects to EMCOMM-NET and reaches the full FieldComms dashboard through a standard web browser in under a minute. No app installation is required. No internet connection is needed for any core feature.

FIELD	VALUE
Wi-Fi Network Name (SSID)	EMCOMM-NET
Password	Provided by your net manager
Security	WPA2
Your device will receive IP	192.168.50.100 – 192.168.50.200

Step 2 — Open the Dashboard

Open any web browser and go to: **http://192.168.50.1**

Step 3 — Identify Yourself

Enter your callsign, Radio ID, or name when prompted. Your identity is saved by your browser and remembered between sessions.

2 Main Dashboard

http://

The dashboard at http://192.168.50.1 is your central hub for every activation. It shows live NWS weather alerts color-coded by severity, a real-time APRS station table fed by Graywolf and YAAC, system health indicators, Dead Man Switch state for any active net, and quick-launch cards for every tool in the system. Select a mode from the three-button mode bar to reorganize the cards for the task at hand.

Three Dashboard Modes

MODE	BUTTON	TOOLS AVAILABLE
Amateur Radio	■	Net Control Logger, Observer Mode, Callsign Lookup, JS8Call, APRS Tactical Map, HF Propagation, Repeaters, Grid Square, NTS Radiogram, ICS-213, ICS-214, Dead Man's Switch, Roster, Pre-Flight, Cheat Sheets, Print Center, Reference Library
Starcom / Public Safety	■	Starcom Net Logger, Weather Net, SAR Net, Observer Mode, Resource Tracking Map, Resource Board, Member Roster, Facilities, ICS Platform, ICS-213, ICS-214, Print Center
ICS / Incident Command	■	ICS Platform (all five sections + Planning P), Tactical Map, Resource Map, Resource Board, Roster, Facilities, ICS forms, Winlink Import, Repeaters, Cheat Sheets

3 Amateur Net Control Logger

http://

The Amateur Net Control Logger is the primary tool for running ARES/RACES nets. Type a callsign and press Enter — FieldComms looks it up in the local FCC database instantly and fills the operator name and license class automatically. Every check-in is timestamped in UTC and saved to the server. Served-agency staff and EOC observers can watch the live log in Observer Mode without touching the net control interface.

Starting a Net

1. Click **[+ New Net]**
2. Enter Net Name → e.g. "McHenry County ARES Thursday Evening Net"
3. Enter Frequency → e.g. "147.015 MHz / 107.2 Hz"
4. Enter Net Control → your callsign
5. Click **[Activate Net]** → The net is now live

Net Control Features

- **FCC Auto-Fill** — Type any US amateur callsign — name and location fill automatically from the local FCC database. No internet required.
- **Multiple Nets** — Run several nets simultaneously. Each has its own log, frequency, and NCS.
- **Observer Link** — Copy a read-only URL for served-agency staff or EOC viewers.
- **ICS-309 Export** — Export the complete net log as a formatted ICS-309 Communications Log.
- **Roster Chips** — Quick-tap chips for known stations. Tap to instantly log a check-in.

4 Starcom Net Logger

http://

Purpose-built for public safety Starcom radio net logging. Identifies units by Radio ID and unit number rather than amateur callsigns, uses the sc- prefix convention for net names.

FEATURE	AMATEUR NET CONTROL	STARCOM LOGGER
Unit identification	FCC Callsign (e.g. W9XYZ)	Radio ID / Unit number (e.g. 1234)
Net name prefix	Free form	sc- prefix (e.g. sc-McHenry-TAC1)
ID lookup	FCC database auto-fill	Unit name / callsign manual entry
Dispatch field	Not applicable	Dispatch center name field included
Export format	ICS-309	ICS-309 (Starcom header)

Starcom nets are distinguished from amateur nets by the sc- prefix. Keep this prefix on all Starcom net names.

5 Net Observer (Read-Only View)

http://

Provides a read-only, auto-refreshing view of any active net. Share the link with served agency personnel, EOC staff, or section leaders who need to monitor traffic without touching the net control interface. No login required.

> ■ The net name is passed in the URL: http://192.168.50.1/observer.html?net=Thursday-Net — The page refreshes automatically every 15 seconds. Bookmark it for quick access.

6 Member Roster

http://

The authoritative directory for McHenry County RACES/ARES/Starcom personnel. Tracks certifications, equipment capabilities, contact information, and deployment activations.

TAB	CONTENTS
Directory	Full member list with callsign, name, role, phone, email, grid square, certification dots, and equipment badges.
Certifications	ICS-100/200/300/400/700/800, EmComm I/II, CPR/AED, First Aid, FEMA IS courses — 11 total.
Equipment	HF/VHF/UHF radio, Winlink, JS8Call, APRS, generator, antenna, laptop, go-kit — 10 capabilities.
Activations	Track which members are deployed, their assigned role, check-in time, and location.
Import/Export	Export roster to CSV for backup. Import from CSV to bulk-load members.

7 Resource Board

http://

Track all personnel, vehicles, and equipment assigned to an activation. Each resource card shows current status and can be cycled through states with a single click.

STATUS	COLOR	MEANING
Available	Green	Resource is on scene and available for assignment
Assigned	Amber	Resource is actively tasked with a specific assignment
Staging	Blue	Resource is en route or at the staging area, not yet deployed
Out of Service	Red	Resource is unavailable — mechanical, medical, or administrative
Demobilized	Grey	Resource has been released from the incident

8 Tactical APRS Map

http://

Displays live APRS station positions on an interactive Leaflet map. Pulls data from Graywolf TNC (port 8080) and YAAC (port 8082), merging them into a unified picture. Works completely offline.

SIDEBAR TAB	CONTENTS
Stations	All heard APRS stations sorted by most recent. Filter by source or search by callsign. Click any station to pan the map.
Messages	Incoming and outgoing APRS messages. Send a message to any callsign via Graywolf or YAAC.
Markers	Place manual overlay markers on the map. Add labels, colors, and notes.

■ Click any station icon on the map to see full detail: APRS symbol, type, comment, speed, course, altitude, path, and last-heard time. The popup also has Track and Message buttons.

9 Starcom Resource Map

http://

A Leaflet-based map for manually placing and tracking Starcom units. Unlike the APRS tactical map, positions here are placed manually by the operator. Supports unit placement, status color coding, zone drawing, and polygon area markup for tactical overlays.

> ■ Use the Starcom Resource Map alongside the Starcom Net Logger during a public safety activation. Place units on the map as they check in, and update their status as assignments change.

10 FCC Callsign Lookup

http://

Search the local FCC amateur license database — the entire US amateur callsign database (~800,000 licensees) stored offline on the Pi. No internet required. Results include licensee name, address, license class, grant date, expiry date, and trustee callsign for club licenses.

> Direct URL lookup: http://192.168.50.1/callsign.html?call=W9XYZ — Useful for bookmarking specific calls or linking from net control exports.

11 Dead Man's Switch Monitor

http://

Monitors each active net for inactivity. If a net goes silent beyond a configured threshold, it escalates through warning and trigger states — alerting operators that the net may have lost communications.

STATE	COLOR	MEANING	ACTION
DISARMED	Grey	Not being monitored	Click ARM to activate monitoring
ARMED	Green	Net is communicating	Normal operations. Reset timer on each check-in.
WARNING	Amber	75% of threshold reached	Contact net control. Resume activity to clear.
TRIGGERED	Red	Threshold exceeded	Immediate action required — check radio and net.

12 Pre-Flight Deployment Checklist

http://

A structured GO / CAUTION / NO-GO readiness assessment to complete before any deployment. Covers all critical systems and saves a timestamped JSON record.

SECTION	ITEMS CHECKED
Power Systems	Shore/generator power, battery backup, UPS, fuel supply, runtime estimate
Communications Equipment	HF radio, VHF/UHF radio, TNC, antennas, coax, SWR check
Computing & Software	Pi boot, all FieldComms services green, Kiwix, FCC DB, browser access
Personnel & Staffing	NCS identified, EC notified, backup NCS available
Field Supplies	Operator cards, printed forms, batteries, cables, tools, PPE

13 ICS Platform — Overview

The ICS Platform provides a complete Incident Command System management interface organized into the five standard ICS sections. All data is saved to the FieldComms server and synchronized in real time across every device on EMCOMM-NET — section chiefs at different workstations all see the same incident state simultaneously without refreshing their browsers. The Planning P tab walks you through all 15 phases of the ICS planning cycle with agendas, required forms, and attendee lists for each phase.

Creating an Incident

1. Open http://192.168.50.1/ics/
2. Click **[+ New Incident]**
3. Fill in Incident Name, Type, Jurisdiction, Incident Commander, Op Period Duration
4. Click **Activate Incident** — all five sections are now live

14 ICS Command

http://

FEATURE	HOW TO USE IT
Incident Objectives (ICS-202)	Enter numbered tactical objectives. Use the 100-item dropdown to select common objectives organized in 13 groups — or type custom objectives directly.
Safety Message	Free-text safety message for the period — weather, hazards, PPE requirements.
Weather Summary	Current conditions and forecast for the operational area.
Command Staff (ICS-203)	Fill in name and agency for each ICS position: IC, Safety Officer, PIO, Liaison Officer, and section chiefs.
Situation Report	Current situation, accomplishments, and planned actions for the IAP.

15 ICS Operations

http://

FEATURE	HOW TO USE IT
T-Card Board	Visual Kanban-style board with four columns: Available / Staged / Assigned / Out of Service. Drag cards to move resources between columns.
Resource List	Table view of all resources with ID, type, name, capability, status, assignment, and contact.
Assignments (ICS-204)	List of all resources in Assigned status with their tasking and contact info.

16 ICS Planning

http://

FEATURE	HOW TO USE IT
Situation Status (ICS-209)	Status summary: total personnel, resources by status, incident narrative, weather, agency list.
IAP Forms	Checklist of all ICS forms with status indicators and links to open each one.
Resource Status Table	Grid showing how many of each resource kind are ordered, on scene, assigned, available, out of service, still needed.
Documentation	Running log of all documents produced during the incident.

17 ICS Logistics

http://:

FEATURE	HOW TO USE IT
Comms Plan (ICS-205)	Build the radio communications plan. Each row is a channel: function, name, frequency, CTCSS tone, mode, remarks. Pre-filled with McHenry County channels.
Supply Tracking	Log supply requests with item, category, quantity needed, quantity on hand, and priority.
Food / Medical (ICS-206)	Meal service log and medical unit leader, ambulance service, nearest hospital info.
Check-In (ICS-211)	Personnel check-in list with name, ICS position, agency, and arrival time.

18 ICS Finance / Admin

http://

FEATURE	HOW TO USE IT
Cost Accounting	Log all expenditures: category, amount, description, vendor, approver. Running total shown.
Time Tracking	Log person-hours for each operator: name, position, agency, hours, hourly rate (0 = volunteer).
Procurement	Track purchase orders: item, vendor, PO number, amount, status.
Export	Export all finance data to CSV for submission to agency finance officer.

19 NTS Radiogram Generator

http://

FIELD	DESCRIPTION	EXAMPLE
Message Number	Sequential message number	001
Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE	ROUTINE
Handling Instructions	HX codes — delivery special instructions	HXC
Station of Origin	Originating station callsign	W9XYZ
Check (word count)	Word count of text — auto-calculated	22
Place of Origin	City and state of originating station	WOODSTOCK IL
Time Filed	UTC time — auto-fill button provided	2145Z

20 ICS-213 General Message & ICS-214 Activity Log

http://

ICS-213 — The standard written message form for inter-agency and intra-agency communications. Fill in the header (incident name, to/from, position/agency, date/time, subject) and the message body. Click **Print** to produce a field-ready paper copy.

> ICS-213 is for formal written communications. For routine spoken radio traffic, use the net control log. Use 213 when you need a written record of requests, situation updates, or resource orders.

ICS-214 — A unit-level activity log required for every ICS section and unit. Records personnel assigned and a timestamped log of activities through the operational period. Click Add Personnel to list team members, then Add Entry for each activity.

21 ICS-309 Communications Log

http://

FIELD	WHAT TO ENTER
Incident Name	Name of the active incident (auto-populated if incident is active)
Operational Period	Current period number and date range
Radio Operator	Your name and callsign
Station/Net	Net name or radio station identifier
Log Rows	Date/Time, From, To, Subject/Message summary — one row per message

Click **Add (timestamp now)** to add a row with the current UTC time. Click **Save to Incident** to file the completed log with the active incident.

22 HF Propagation

http://

INDICATOR	WHAT IT MEANS FOR HF
SFI (Solar Flux Index)	Higher = better high-band propagation. >130: excellent. 90–130: good. <90: 40m/80m only.
Sunspot Number	More sunspots = better HF. Follows 11-year cycle.
A-Index	Daily geomagnetic activity. <10 quiet (good HF). 10–30 unsettled. >30 disturbed.
K-Index (0–9)	3-hour snapshot. ≤2 quiet. 3–4 unsettled. ≥5 storm (HF degraded). ≥7 severe.
Band Condition Bars	Green=good / Amber=fair / Red=poor for each band from 10m through 80m.

23 Repeater Database

http://

SOURCE TAB	HOW TO USE
Offline File (recommended)	Load a RepeaterBook CSV or JSON export. Drag-and-drop or Browse. Data persists in the browser after first load.
Server API	Pulls from FieldComms server if fetch_repeaters.py has been run. Requires RepeaterBook API token.
Sample Data	Demo repeaters for testing the interface. Not real — do not use for operations.

Filters: **Band** (2m, 70cm, etc.) · **State** · **Affiliation** (ARES, SKYWARN, RACES) · **Mode** (FM, DMR, C4FM, D-STAR) · **Sort** by callsign, frequency, city, or distance

24 Facilities Directory

http://

Maintain a directory of EOC locations, hospitals, shelters, staging areas, and command posts. Each entry stores address, coordinates, radio frequencies, CTCSS tone, contact person, on-site callsign, generator status, ADA access, capacity, and operational notes.

> ■ Four default facilities are pre-loaded for McHenry County. Edit these with your actual EOC, shelter, and staging area information. Click any card to expand full detail.

25 Grid Square Calculator

http://

Convert between decimal latitude/longitude and Maidenhead grid squares in both directions. Supports 4-character (EN52) and 6-character (EN52ab) precision. Also calculates distance and bearing between any two grid squares. McHenry County, IL is in grid square **EN52**.

26 Radio Cheat Sheets

http://

TAB	CONTENTS
Phonetic Alphabet	NATO phonetics A–Z with pronunciation, plus ITU number pronunciation
Q-Codes	Common amateur Q-codes and EmComm net Q-codes (QNI, QNS, QND, QNN, etc.)
Prowords	ROGER, WILCO, SAY AGAIN, BREAK, OVER, OUT, CORRECTION, SILENCE, and 25 others
Band Plan	2m and 70cm segments, HF emergency frequencies, MURS, FRS, GMRS, Marine, Aviation
NTS Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE — definitions and when to use each
ICS Positions	Standard ICS command and general staff titles with abbreviations
CTCSS / DCS	Standard CTCSS tone table (67.0–254.1 Hz) and DCS code table

27 Print Center

http://

CATEGORY	AVAILABLE DOCUMENTS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, NTS Radiogram, Pre-Flight Checklist
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, CTCSS/DCS & Signal Reports
Operations	Net Control Log (ICS-309), Starcom Net Log, Member Roster, Resource Board
Cover Sheet Generator	Fill in incident details → generates formatted IAP cover sheet
Operator Access Cards	Avery 5371 format — 10 cards per sheet — callsign, EMCOMM-NET, IP address

28 Kiwix Offline Library

http://

CONTENT	WHAT'S IN IT	BEST FOR
WikiMed Medical Encyclopedia	Symptoms, diagnoses, treatments, medications, procedures, anatomy	Mass casualty events, medical support operations
Wikipedia (Mini)	Compact English Wikipedia — facts, geography, science, history	General field reference, briefing preparation
Wikivoyage	Travel guides, emergency shelters, local resources, evacuation routes	Shelter-in-place planning, unfamiliar jurisdictions
iFixit Repair Guides	Equipment repair procedures for electronics, tools, vehicles	Field equipment troubleshooting

29 Health Monitor

http://

METRIC	NORMAL RANGE	ACTION IF EXCEEDED
CPU Temperature	< 70°C	Improve ventilation. Pi 5 throttles at 85°C.
Memory Usage	< 80%	Close unused browser tabs. Restart non-essential services.
Disk Usage	< 90%	Archive old logs. Remove unused ZIM files.
Service Status	All green dots	Red dot = service down. Restart with: sudo systemctl restart SERVICE
Internet	Connected (when available)	Check Ethernet. All core features work offline.

30 Quick Reference & Tips

TOOL	URL	WHEN TO USE
Dashboard	/	Opening screen — system status, weather alerts, APRS, tool access
Net Control	/netcontrol.html	Running any ARES/RACES amateur radio net
Starcom Logger	/starcom.html	Public safety Starcom nets with Radio IDs
ICS Platform	/ics/	Any formally-declared ICS incident
Tactical Map	/tactical.html	APRS situational awareness, unit tracking
Pre-Flight	/preflight.html	Before every activation — GO/CAUTION/NO-GO
Roster	/roster.html	Member management, activation tracking
NTS Radiogram	/nts.html	Generating and logging formal ARRL traffic
Kiwix Library	:8081/	Medical reference, equipment repair, field skills
Health Monitor	:5051/health	Check service status, system temperature, disk usage

■ During an activation, the Health Monitor at http://192.168.50.1:5051/health shows complete system status including CPU temperature, memory, disk usage, service health dots, GPS fix quality, and internet connectivity. If a service dot is red, SSH to the Pi and run: sudo systemctl restart . Common service names: fcc-lookup, health-monitor, ics-platform, kiwix, deadmans, nginx. The install log is at /var/log/fieldcomms-install.log.

31 Reference Library

http://

Stores all field reference documents — radio manuals, interoperability plans, ICS form packages, SOGs, agency plans, and training materials. Files stored on the Pi, accessible from any device on EMCOMM-NET.

- Click ■ **Upload Document**. A panel slides in from the right.
- Drag-and-drop the file or click Browse. Accepts PDF, Word, Excel, images, KML, ZIP, GPX — up to 200 MB.
- Fill in Title, Category, Source, Description, Tags, Revision, and Expiry.
- Click ■ **Upload Document**. PDFs get an automatic thumbnail.

32 Winlink Form Import & Incident Archiving

http://:

Brings ICS form data from Winlink Express into the FieldComms incident record. When Winlink Express sends or receives an ICS form, it attaches an XML data file. This page parses that XML, maps the fields, and archives the form to the active incident.

Step-by-Step Import

1. In Winlink Express, open the ICS form message (sent or received).
2. Right-click the **RMS_Express_Form_*.xml** attachment and save it.
3. On the Winlink Form Import page, drag the saved XML file onto the drop zone, or click Browse.
4. Click **Parse Form Data**. The page detects the form type and extracts all fields.
5. Review the extracted fields, select the Incident, then click **Archive to Incident**.

33 JS8Call — HF Digital Keyboard Messaging

JS8Call is a keyboard-to-keyboard HF digital mode based on FT8 that provides store-and-forward messaging, heartbeats, and directed messaging at very weak signal levels. Runs on the Windows laptop connected to the IC-7300.

Setup

- JS8Call must be running on the Windows laptop (not the Pi).
- Enable the API: File → Settings → API — enable, set port to **2442**.
- Set hostname to **0.0.0.0** to allow EMCOMM-NET devices to connect.
- Note the Windows laptop IP address on EMCOMM-NET (e.g. 192.168.50.105).
- On FieldComms dashboard, tap the JS8Call card → enter the laptop IP → tap OK.

34 ICS Planning P — Operational Planning Cycle

http://

Interactive guide to the ICS operational planning cycle. Presents all 15 phases from incident through each operational period, with the standard agenda, required ICS forms, and attendees.

COLOR	GROUP	PHASES
Gray	Initial Response — incident through incident brief	Phases 1–5
Yellow	Establish Objectives — IC/UC objectives + C&G; Staff Meeting	Phases 6–7
Red	Develop the Plan — tactics through planning meeting	Phases 8–11
Green	Prepare & Disseminate — IAP prep + operations briefing	Phases 12–13
Teal	Execute, Evaluate & Revise — execute plan + new ops period	Phases 14–15

Click any phase button → see the standard agenda, required forms (as clickable links), and attendees. Click **Generate briefing sheet** to print a one-page cover for the IAP.

35 Operator Access Cards

http://

Printable Avery 5371 business-card-size reference cards (2" × 3.5", 10 per sheet). Each card shows the network name, server IP address, and quick-reference steps. Print, cut, laminate, and give one to every operator at a deployment.

CARD FIELD	CONTENT
Network	EMCOMM-NET (Wi-Fi name)
Server address	http://192.168.50.1
Quick steps	Connect → Open browser → Enter identity
Callsign / ID	Member callsign (large, blue) or Radio ID (large, purple)
ESV Member ID + Radio ID	Shown on all cards regardless of type

36 Network — EMCOMM-NET & AiMesh Coverage Extension

EMCOMM-NET is the Wi-Fi network broadcast by the ASUS RT-BE58 Go travel router. All FieldComms tools are accessible at http://192.168.50.1 from any device connected to EMCOMM-NET. The Pi does not run its own Wi-Fi.

OPTION	HOW IT WORKS	BEST FOR
Single router	RT-BE58 Go covers ~2,000–2,500 sq ft. No additional setup needed.	Single room EOC activations
AiMesh nodes	Add ASUS AiMesh nodes to extend coverage. All nodes share EMCOMM-NET SSID and 192.168.50.x subnet. Devices roam automatically.	Multi-room EOC buildings, large shelters, outdoor staging areas

- Factory reset the node router. Hold reset 5–10 seconds.
- Power on the node within 30 ft of the primary router for pairing.
- On primary router: **AiMesh** → **Add AiMesh Node**. Select the node and connect.
- Move the node to its final position once pairing completes (~3 minutes).

37 Printer Setup

http://1

FieldComms has print buttons on 17 pages — all using the browser standard print function. Three connection options are available:

OPTION	HOW IT WORKS	BEST FOR
A — Own printer	Each operator prints to their own locally connected printer. No Pi setup needed — works immediately.	Single operator or when every device already has a printer
B — USB printer via Pi (CUPS)	USB printer plugged into the Pi. CUPS shares it across EMCOMM-NET. Admin at http://192.168.50.1:631. Installed automatically by the FieldComms installer.	Field activations — one shared printer for all operators. Supports Windows, Mac, iOS AirPrint, Android, Chromebook.
C — Network printer	Wi-Fi or Ethernet printer connected directly to EMCOMM-NET. Devices discover it via Bonjour/mDNS automatically.	Sites with an existing Wi-Fi capable printer

- Plug the USB printer into the Pi or powered USB hub.
- Open **http://192.168.50.1:631** from any device on EMCOMM-NET.
- Click **Administration** → **Add Printer** and log in with Pi username/password.
- Select printer from Local Printers. Check **Share This Printer**. Click Continue.
- Select driver, click Add Printer, print a test page.

Recommended field printers: Brother HL-L2350DW (laser, excellent Linux support), Canon PIXMA TR150 or HP OfficeJet 200 (battery-powered portable options).

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	**Cortex-A76 @ 2.4 GHz (Pi 5 generation)**	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	**8 GB LPDDR4X**	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	**TBD (current model)**	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	**✓ YES — Pi 5 chip, 8 GB RAM, current generation**	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

■ The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit ($120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit ($120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

■ **501(c)(3) Grant Funding Note:** MCESV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	**Pi 500 + Pi Monitor**	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed

■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.